

Summary

* Shift Cipher

• basically moves every plaintext letter by the same fixed amount.

• encryption $\Rightarrow$ $c = E(p) = (p + k) \pmod{26}$

• decryption $\Rightarrow$ $p = D(c) = (c - k) \pmod{26}$

p : plain text, k : shift key, c : ciphertext value

• this can be easily broken by trying out all the 26 possible shifts
- to basically brute force

* Affine Cipher

• Here, we apply both addition & multiplication

• Encryption $\Rightarrow$

$$c = E(p) = (ap + b) \pmod{26}$$

equivalent to linear equation: $y = ax + b$

& to be reversible: $\gcd(a, 26) = 1$

else, unique decryption is impossible

• Decryption $\Rightarrow$

$$c \equiv ap + b \pmod{26}$$

$$c - b \equiv ap \pmod{26}$$

$$p = D(c) = a^{-1}(c - b) \pmod{26}$$

* Chosen plaintext & chosen ciphertext attacks

- * In a chosen plaintext attack, the attacker can select plaintext & observe the resulting ciphertext. Carefully selected plaintext letters can simplify the equations and allow the cipher to be recovered.
- * In a chosen ciphertext attack, the attacker selects ciphertext and observes the resulting plaintext. The resulting equations can be used to recover parameters of the decryption function and then determine the encryption key.

* Vigenere Cipher

- * This works similarly to a shift cipher, but it uses a sequence of shifts rather than one fixed shift.
- * A keyword is converted to numerical values: $k_1, k_2, \dots, k_m$
- * if $\text{len}(\text{plain text}) > \text{len}(\text{key}) \Rightarrow$ repeat it from the beginning.
- * Encryption $\Rightarrow$ $C_i = (P_i + K_i) \pmod{26}$
- * Decryption $\Rightarrow$ $P_i = (C_i - K_i) \pmod{26}$
- * However, if the key is short & repeated a lot, patterns are introduced. These might then be recovered by cryptanalysis.

✧ XOR encryption $\Rightarrow$

✧ encryption works bit by bit $\Rightarrow$ $C_i = P_i \oplus K_i$

✧ it's equivalent to $\rightarrow C_i = (P_i + K_i) \pmod{2}$

✧ & same operation does both encryption & decryption $\Rightarrow$ $C = P \oplus K$
 $P = C \oplus K$

✧ this works because $K \oplus K = 0$ & $P \oplus 0 = P$.

✧ Permutations

✧ just rearrange the position of the letter accordingly.

✧

A	B	C	D
1	2	3	4

$\leftarrow$ positions

for permutation key $(3, 1, 4, 2) \Rightarrow C A D B$

✧ Substitution Cipher

- ✧ replaces each plaintext symbol with another symbol
- ✧ mappings are very unique, not repeated at all, if so, will break
- ✧ follows a pre-defined set of substitutions/rules.
- ✧ basically like a 1-1 permutation of the alphabet.
- ✧ these are not secure & can be cracked with statistical analysis

key

$A \rightarrow Z, B \rightarrow P$

$\Rightarrow$

example

$ABCD \Rightarrow ZPMZ$

$C \rightarrow M, D \rightarrow Z$

$\leftarrow$ by referring to the key

• Block Cipher

- basically split the thing into fixed lengthed blocks & do the thing
- instead of dealing with individual letters
- block sizes are based on powers of two
- eg: hill cipher

• Hill Cipher

- We use an invertible matrix as the encryption key
- for an $n \times n$ matrix, we divide the plaintext in n letters.
- each block is then $(\text{mod } 26)$ to bring values to 0-25.

• Encryption $\Rightarrow C = P \times M \pmod{26}$

P: PT. row vector, M: matrix key, C: CT. row vector

- if column vectors are used, we change the multiplication order
- if PT is not divisible to blocks properly, add padding.

• Example $\Rightarrow \underline{HI} \mid \underline{RU}, M = \begin{bmatrix} 1 & 2 \\ 4 & 7 \end{bmatrix}$

$$HI \Rightarrow [7 \ 8]$$

$$RU \Rightarrow [17 \ 20]$$

$$\begin{bmatrix} 1 & 2 \\ 4 & 7 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 \\ 4 & 7 \end{bmatrix}$$

$$[7 \ 8] \begin{bmatrix} 13 & 18 \end{bmatrix} \Rightarrow NS$$

$$[17 \ 20] \begin{bmatrix} 19 & 18 \end{bmatrix} \Rightarrow TS$$

$$\therefore HIRU \Rightarrow NSTS$$

* Decryption $\Rightarrow$

$$P = C \times M^{-1} \pmod{26}$$

- * only possible when $\gcd(\det(M), 26) = 1$.
- * if gcd is not 1, that matrix X be used for unique decryption

* inverse of a 2×2 matrix $\Rightarrow$

$$M = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \quad \det(M) = ad - bc$$

$$\text{and } M^{-1} = (\det M)^{-1} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \pmod{26}$$

* inverse of any larger matrix is

$$M^{-1} = (\det M)^{-1} \text{adj}(M) \pmod{26}$$

$$\text{where } \text{adj}(M) = [C(M)]^T$$

* Recovery an unknown Matrix $\Rightarrow$

* need to know both PT (matrix A) and CT (matrix B)

$$A \times M = B \pmod{26}$$

if A is invertible

$$A^{-1} \times A M = A^{-1} B$$

therefore

$$M = A^{-1} B \pmod{26}$$

also, PT must satisfy $\gcd(\det A, 26) = 1$

otherwise, A^{-1} does not exist

• Example $\Rightarrow$

S	I	T	C	R	Y	P	T	O
C	I	H	O	L	I	N	F	O

3x3 matrix

$$SIT \Rightarrow [18 \ 8 \ 19]$$

$$CRY \Rightarrow [2 \ 17 \ 24]$$

$$PTO \Rightarrow [15 \ 19 \ 14]$$

$$\therefore A = \begin{bmatrix} 18 & 8 & 19 \\ 2 & 17 & 24 \\ 15 & 19 & 14 \end{bmatrix}$$

$$CIH \Rightarrow [2 \ 8 \ 7]$$

$$OLI \Rightarrow [14 \ 11 \ 8]$$

$$NFO \Rightarrow [13 \ 5 \ 14]$$

$$\therefore B = \begin{bmatrix} 2 & 8 & 7 \\ 14 & 11 & 8 \\ 13 & 5 & 14 \end{bmatrix}$$

$$A \times M = B$$

$$\times A^{-1} \quad \underbrace{A^{-1} A} M = A^{-1} B$$

$I \Rightarrow$ the identity matrix

$$M = A^{-1} B$$

$$\text{find } C(A) \Rightarrow C(A) = \begin{bmatrix} 16 & 20 & 17 \\ 15 & 19 & 12 \\ 25 & 22 & 4 \end{bmatrix}, \text{ by solving } C_{11}, C_{12}, C_{13}, C_{21}, \dots, C_{33}$$

$$\text{find } \text{Adj}(A) = \begin{bmatrix} 16 & 15 & 25 \\ 20 & 19 & 22 \\ 17 & 12 & 4 \end{bmatrix}$$

$$\text{find } \det(A) = 18C_{11} + 8C_{12} + 19C_{13} = 771 \pmod{26} = 17$$

find 17^{-1} with extended euclidean algorithm

$$\Rightarrow 1 = 2 \times 26 - 3 \times 17$$

$$\therefore -3 \times 17 = 1 \pmod{26} \Rightarrow 17^{-1} \equiv -3 \pmod{26}$$

$$\equiv 23 \pmod{26}$$

$$\therefore A^{-1} = 23 \times \text{Adj}(A) \pmod{26}$$

$$A^{-1} = \begin{bmatrix} 4 & 7 & 3 \\ 18 & 21 & 12 \\ 1 & 16 & 14 \end{bmatrix} \pmod{26}$$

$$\text{and } M = A^{-1} \times B = \begin{bmatrix} 15 & 20 & 22 \\ 18 & 19 & 20 \\ 18 & 20 & 19 \end{bmatrix} \pmod{26}$$

regular matrix multiplication